

Question ID: 5dce6cab

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	Hard

Question

Given that the conditions in binary star systems should make planetary formation nearly impossible, it’s not surprising that the existence of planets in such systems has lacked _____ explanation. Roman Rafikov and Kedron Silsbee shed light on the subject when they used modeling to determine a complex set of factors that could support planets’ development.

Which choice completes the text with the most logical and precise word or phrase?

- Answer**
- A. a discernible
 - B. a straightforward
 - C. an inconclusive
 - D. an unbiased

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s description of efforts to explain the existence of planets in binary star systems. As used in this context, describing an explanation as “a straightforward” one would mean that the explanation is direct and uncomplicated. The text asserts that since it should be “nearly impossible” for planets to form in binary star systems, it’s “not surprising” that there isn’t a straightforward explanation for the existence of planets in such systems; the fact that one potential approach involves “complex” factors offers further contextual support for this idea.

Choice A is incorrect because it would not make sense in context to say that there isn’t “a discernible” explanation—meaning an explanation capable of being perceived—for the existence of planets in binary star systems. The text discusses just such an explanation offered by Roman Rafikov and Kedron Silsbee, which indicates that their explanation can be discerned. Choice C is incorrect because the text emphasizes how difficult it is to explain the existence of planets in binary star systems, suggesting that the situation isn’t marked by the lack of “an inconclusive” explanation—an explanation that does not resolve the issue—but rather that if any explanations have been offered, they’ve likely been inconclusive ones. Choice D is incorrect because nothing in the text suggests that there is a lack of “an unbiased,” or impartial and unprejudiced, explanation for the existence of planets in binary star systems. The text indicates that it’s difficult to explain the existence of planets in such systems and it describes one attempt to do so, but there is no evidence that explanations from Roman Rafikov and Kedron Silsbee or others are biased.